

Andre Tran

+1 (513) 356-1202 | anhdtran.forwork@gmail.com | linkedin.com/in/anhdtran11 | github.com/trnahnh | anhdtrn.com

EDUCATION

University of Cincinnati

Bachelor of Science in Computer Engineering, Minor in Computer Science

Expected May 2028

GPA: 3.71/4.00

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design & Development, Cybersecurity, Digital Design & Digital Systems, Artificial Intelligence, Context Engineering

Affiliated: CodePath, University of Cincinnati Powerlifting Team, University of Cincinnati Bodybuilding Club

EXPERIENCE

Commma - Pace your Code (commma.dev)

05/2026 - Present

Founder & Lead Engineer

Remote

- Engineering commma.dev end-to-end: a full-stack TypeScript monorepo with a published editor extension (VS Code + Open VSX, 200+ installs) capturing per-keystroke telemetry and keyboard heatmap export.
- Processing 271K+ events into sessions via a distributed aggregation pipeline, sustaining zero 5xx errors and read p95 under 200ms at 100 concurrent users across 33K+ total requests in production.
- Replacing BullMQ with an in-process aggregator to eliminate idle Redis polling; engineering distributed deduplication via Redis leader lock (SET NX PX) and cutting PostgreSQL privacy lookups from 42ms to 13ms p50.
- Provisioning AWS infrastructure as code in Terraform (EC2 t4g, S3 + CloudFront, Route 53) with zero downtime; automating path-filtered GitHub Actions CI/CD and wiring Grafana Alloy to Loki for 5 production SLOs.

Caphne - Study Buddies Matchmaking (caphne.co)

01/2026 - Present

Lead Software Engineer

Remote

- Engineering a distributed real-time messaging system via Socket.IO, PostgreSQL, and Redis, reducing API response time from 60s to 10s and sustaining 1.7K+ requests/minute over 30 days of traffic.
- Leading a cross-functional team of 6 engineers through sprint planning, code reviews, and production deployments in Nuxt 3 and Node.js, scaling the study-matching platform to 400+ active users.
- Architecting Vue frontend with lazy loading and input debouncing; migrating to AWS (EC2 t4g, S3 + CloudFront), achieving 4x faster page loads with p50 under 400ms and cutting server queries by 40%.

KatanaID - AI Branding Toolkit (katanaid.com)

12/2025 - Present

Founding Engineer

Remote

- Leading 5 engineers to architect a Go backend leveraging Goroutines to orchestrate 19+ external API calls for real-time brand asset aggregation and PDF generation.
- Handling 2.3K+ requests/day under production load (k6-validated) with Google Gemini AI for brand asset generation alongside a trust score engine with browser fingerprinting.
- Eliminating schema-related runtime errors across 19+ API integrations via Ent ORM's strictly type-safe PostgreSQL layer, reducing production incidents from schema drift to zero.

PROJECTS

Draft-Thinker (Live) | Go, OpenAI API, Qdrant, Redis, Prometheus, Grafana, Docker

- Reduced LLM inference costs by 91.6% by routing 94% of requests through a drafter model via real-time Shannon entropy analysis of token logprobabilities, achieving 98.2% accuracy on a 518-prompt benchmark.
- Implemented a semantic cache layer with vector search and TTL-based eviction serving repeated queries under 50ms at cosine similarity above 0.95, eliminating redundant inference on cached draft-accepted responses.

Ferrox (Live) | Rust, Systems Programming, Atomics, Mmap, Criterion

- Achieved 500ns P99 tick-to-trade latency (verified via HdrHistogram) processing 4.7M simulated orders/second by building a matching engine with zero hot-path allocations via a 1M-slot arena and doubly linked lists.
- Increased throughput by 8.8x by engineering a lock-free SPSC ring buffer with Acquire/Release memory ordering, implementing cache-line padding (64B) and a mmap write-ahead log for crash recovery under 1.4ms.

TECHNICAL SKILLS

Languages: Go, JavaScript/TypeScript, SQL, Python, Rust, C/C++, C#, Java, HTML/CSS

Technologies: Nuxt.js, Vue.js, Next.js, React, Node.js, Bun, Express.js, Spring Boot, Chi, Gin, Hono, AWS (Amazon Web Services), Docker, PostgreSQL, MongoDB, MySQL, FastAPI, Django, Flask, LangChain/LangGraph, ChromaDB, scikit-learn, NumPy, Socket.IO, Git, GitHub, Grafana, Prometheus, Redis, Terraform, CI/CD, Linux